

PIYUSH RAJ

B.Tech Agricultural Engineering (Final Year) | CATIA V5 | Tractor & Agricultural Machinery Design

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Professional summary

Final-year B.Tech Agricultural Engineering student with hands-on experience in tractor systems, CATIA V5 3D modeling, assembly design, and engineering drafting. Interned at CFMTTI Budni and ICAR-IISWC Dehradun, gaining exposure to hydraulic systems, PTO mechanisms, transmission systems, tractor testing, and preventive maintenance. Proficient in CATIA V5, ANSYS, GD&T, and machine design.

Education

Centurion University of Technology and Management (CUTM), Odisha

B.Tech in Agricultural Engineering | CGPA: **8.61** | Expected Graduation: **2027**

Technical skills

CAD: CATIA V5, SolidWorks, 3DEXPERIENCE

Analysis: ANSYS

Engineering: GD&T, Machine Design, Engineering Drawing, Tractor Systems

Design: 3D Part Modeling, Assembly Modeling, Drafting, BOM Preparation

Programming: Python (Basic)

Internship experience

CFMTTI, Budni (2026) – Tractor Systems & Testing Intern

Studied engine, transmission, hydraulic, PTO, steering, and braking systems; assisted in inspection, preventive maintenance, and performance testing; prepared technical reports.

ICAR-IISWC, Dehradun (2025) – Agricultural Engineering Intern

Worked on soil and water conservation, irrigation, and watershed management projects; assisted in technical surveys and engineering documentation.

Key projects

Tractor Component Design Using CATIA V5: Designed 9 tractor components with complete 3D models, assemblies, and engineering drawings, including 3-Point Hitch Assembly, Lift Arms, Wheel Hub, Gearbox Cover, Clutch Box, and Front Wheel Stub Axle Spindle.

Battery-Powered Lawn Mower: Designed a prototype integrating a DC motor drive, battery power system, wheel assembly, and cutting mechanism with a focus on lightweight design, energy efficiency, and maintainability.

Certifications & leadership

Certifications: NPTEL – Micro Irrigation Engineering; CATIA V5 Design Practice; Tractor Systems and Maintenance Training.

College Secretary, CUTM: Coordinated technical events, student activities, and team communication.